

Assignment-02

Design and Implementation of a Multi-Step Prompt Workflow for Automated Interview Preparation and Personalized Mock Interview Generation

AI-Based Resume Screening System for Automated Candidate Evaluation

Submitted by:

Name: KANAPARTHI JAYAVARDHAN REDDY

Registration No: 99230040324

Programme: GEN-AI

Module: Prompt Engineering

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1. Introduction

Recruitment teams often receive a large number of resumes for a single job opening. Manually reviewing every resume can require significant time and may lead to inconsistent screening decisions. An AI-based resume screening system can help organize this process by extracting relevant information from resumes and comparing candidate profiles with job requirements.

This assignment presents a structured approach for an AI-based resume screening system. The workflow separates the process into three connected stages: resume and job description analysis, candidate matching and scoring, and final review and recommendation. The output of each stage is used by the next stage so that candidate evaluation remains structured and easier to validate.

2. Objective

The main objective is to design a reliable workflow that converts resume information and job requirements into structured candidate evaluation results. The workflow aims to identify candidate qualifications, compare them with job requirements, generate a structured matching score, and apply a review stage to improve consistency.

3. System Design

The proposed system treats resume screening as a pipeline of three specialized stages. Stage 1 analyzes the resume and job description. Stage 2 matches candidate information against the identified requirements and produces a structured score. Stage 3 validates the result and creates a final recommendation.

Stage 1 – Analysis: Extracts candidate information and identifies job requirements.

Stage 2 – Matching: Compares skills, education, experience, and relevant projects.

Stage 3 – Review: Checks the evaluation and produces the final candidate recommendation.

4. Methodology

The workflow is developed using three connected processing stages. Each subsequent stage receives the output of the previous stage. This creates a controlled pipeline in which the final recommendation is based on the information and criteria established during the earlier stages.

4.1 Step 1: Resume and Job Description Analysis

The first stage converts the resume and target job description into structured information. It identifies the candidate's education, technical skills, programming languages, projects, experience, certifications, and other relevant qualifications.

Prompt 1: Act as an expert recruitment analyst. Analyze the candidate resume and the target job description. Extract the candidate's education, technical skills, programming languages, projects, experience, certifications, and other relevant qualifications. Identify the key requirements of the job description. Organize the information into a structured candidate profile and job requirement summary.

4.2 Step 2: Candidate Matching and Scoring

The second stage compares the structured candidate profile with the job requirements. The system evaluates relevant skills, education, experience, and projects and produces a matching score along with strengths and gaps.

Prompt 2: Using only the candidate profile and job requirement summary generated in Step 1, evaluate the candidate's suitability for the target role. Compare the candidate's skills, education, projects, and experience with the required qualifications. Generate a structured matching score, list the strongest matches, identify important gaps, and provide a concise explanation.

4.3 Step 3: Review, Validation and Recommendation

The third stage acts as a quality-control step. It checks whether the matching result is supported by the resume and job description, whether important requirements were considered, and whether the recommendation is logically consistent.

Prompt 3: Act as a strict senior recruitment reviewer. Review the candidate evaluation for requirement coverage, evidence from the resume, score consistency, skill gaps, strengths, weaknesses, and fairness of the final recommendation. First list any issues found. Then provide the corrected final candidate evaluation.

5. Results and Discussion

The AI-based resume screening workflow provides a systematic approach to candidate evaluation. The analysis stage creates a structured representation of both the candidate and the job requirements before matching begins. This helps reduce the possibility of overlooking important information.

The matching stage provides controlled comparison between the candidate profile and the target role. The system can identify specific skills that match the role and highlight areas where the candidate may require additional knowledge or experience.

The final review stage provides an additional safeguard. It can identify unsupported claims, missing requirements, inconsistent scores, or recommendations that do not match the evidence in the resume. This makes the final evaluation more structured and easier for a recruiter to review.

Overall, dividing resume screening into separate analysis, matching, and review stages improves controllability and makes the evaluation process easier to understand.

6. Conclusion

The design of an AI-based resume screening system demonstrates how structured prompt workflows can be applied to recruitment and candidate evaluation. Instead of depending on a single prompt, the proposed approach uses three cooperating stages: resume and job analysis, candidate matching, and final validation.

The workflow creates clear checkpoints for extracting candidate information, identifying job requirements, evaluating the level of match, and validating the final recommendation. It can be adapted to different job roles by changing the target job description and evaluation criteria.

Therefore, a structured AI-based resume screening workflow provides a practical and repeatable method for organizing candidate evaluation while improving consistency, transparency, and controllability.